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SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

NutriTrace

A Full-Stack Data Engine for Global Food Processing and Additive Risk Analytics

A Capstone Project Report

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Abstract

Modern consumers are exposed to an increasingly complex food supply chain in which packaged products contain chemical preservatives, synthetic colourants, and other additives whose long-term health effects are poorly understood by the average buyer. Existing nutrition applications largely limit themselves to counting macronutrients and comparing barcodes against static databases, leaving a significant gap in ingredient-level chemical transparency and cross-border regulatory awareness. NutriTrace is proposed as a full-stack data engineering solution that ingests raw food packaging images, extracts structured ingredient and additive information using Optical Character Recognition (OCR) and Natural Language Processing (NLP), and cross-references this data against multiple international regulatory frameworks — the United States Food and Drug Administration (FDA), the European Food Safety Authority (EFSA), and the Food Safety and Standards Authority of India (FSSAI).

The system further computes a personalised Acceptable Daily Intake (ADI) for each detected additive based on the user's age and body weight, drawing on World Health Organization / Joint FAO-WHO Expert Committee on Food Additives (WHO/JECFA) thresholds, and applies a rule-based biochemical interaction engine to flag potentially harmful food pairings. A cumulative monthly tracker allows the system to monitor long-term additive exposure and surface early warnings before consumption patterns become a health risk. This report documents the problem motivation, objectives, literature review, comparison with existing systems, proposed architecture, and implementation plan for the NutriTrace platform, and concludes with a discussion of its expected impact and directions for future work.

1. Introduction

Public awareness of nutrition has traditionally centred on calorie counts and macronutrient breakdowns — protein, carbohydrate, and fat content — while the chemical composition of processed foods has remained comparatively opaque to the everyday consumer. Ultra-processed foods routinely contain synthetic dyes, preservatives, emulsifiers, and flavour enhancers identified only by cryptic E-numbers on a label, and few consumers have the technical background or the time to look up what each additive is, whether it is restricted anywhere in the world, or how much of it is safe to consume regularly. This gap between raw label information and actionable health knowledge is the central problem that the NutriTrace project addresses.

1.1 Background and Motivation

Regulatory bodies across the world do not apply uniform standards to food additives. An additive that is approved for use by the FDA in the United States may be restricted or banned outright by the EFSA in the European Union or by the FSSAI in India, reflecting differences in risk tolerance, scientific review processes, and regional dietary patterns. A consumer travelling internationally, or simply purchasing imported packaged goods, has no simple way of knowing whether a product that is considered safe in one jurisdiction is viewed very differently elsewhere. NutriTrace is motivated by the observation that this regulatory fragmentation, combined with the sheer difficulty of parsing dense ingredient lists, leaves an information gap that a well-designed data engineering pipeline is well positioned to close.

1.2 Problem Statement

Existing consumer-facing nutrition tools largely stop at counting calories and macronutrients, and where they do attempt ingredient-level analysis, they typically rely on barcode lookups against a single regional database. This leaves several problems unresolved:

- Hidden chemical preservatives, synthetic dyes, and markers of ultra-processing are not surfaced to the user in an interpretable way.
- There is no mechanism to compare a single additive's legal and safety status across multiple regulatory jurisdictions simultaneously.
- Safety thresholds presented to users are generic rather than personalised to the individual's age and body weight.
- Food items are evaluated in isolation, so dangerous combinations — such as pairing a highly acidic beverage with a protein-dense meal — go undetected.
- There is no cumulative, longitudinal record of a user's chemical additive exposure over time.

1.3 The NutriTrace Approach

NutriTrace is conceived as an end-to-end pipeline that begins the moment a user photographs a food label. A vision layer performs OCR and NLP-based parsing to extract ingredient names and E-numbers from what is often noisy, unstructured text. This structured output is then passed through a multi-jurisdictional regulatory intelligence layer that checks the additive against FDA, EFSA, and FSSAI datasets simultaneously, flagging any discrepancy in legal status. A toxicology module computes the user's personalised ADI using WHO/JECFA reference thresholds together with the user's declared age and body weight, while a separate biochemical interaction engine checks for known adverse food pairings. Finally, a monthly chemical tracker accumulates this information over time, giving the user an early warning before a pattern of consumption becomes a genuine health concern.

1.4 Scope of the Project

The scope of this capstone project covers the design and prototyping of the full NutriTrace data pipeline: image-based ingredient extraction, multi-jurisdictional regulatory mapping, personalised risk scoring, biochemical interaction detection, and longitudinal exposure tracking. The project focuses on packaged and processed foods sold with printed ingredient labels, and on the three regulatory bodies identified as primary reference points — the FDA, the EFSA, and the FSSAI — with the architecture designed to be extensible to additional jurisdictions in future iterations. Fresh, unpackaged produce and restaurant-prepared meals, which do not carry a standardised ingredient label, fall outside the current scope but are noted as a candidate direction for future work.

2. Objectives

The NutriTrace project is organised around four core technical objectives, each targeting one stage of the overall data pipeline.

2.1 Automated Ingredient Extraction

The first objective is to ingest unstructured text captured from scanned packaging images and to clean, normalise, and extract chemical additives, E-numbers, and nutritional values into a structured, machine-readable format. This requires robust OCR tuned to the small, dense fonts typical of ingredient panels, followed by an NLP normalisation stage that resolves spelling variants, abbreviations, and synonyms into a canonical additive identifier.

2.2 Multi-Jurisdictional Regulatory Data Mapping

The second objective is to construct an integrated, cross-border data pipeline that maps each extracted additive against the FDA, EFSA, and FSSAI safety datasets. The pipeline highlights regional usage limits, safety variances between jurisdictions, and country-specific bans, giving the user a single consolidated view of an additive's global regulatory standing rather than a single-country snapshot.

2.3 Statistical Risk Modeling and ADI Calculation

The third objective is to design a mathematical scoring framework — the Additive Risk Index (ARI) — together with an Acceptable Daily Intake (ADI) calculation, in order to statistically quantify safe consumption frequencies. Both calculations are personalised using the user's body weight and age, in line with WHO/JECFA methodology, rather than presenting a single generic threshold to every user.

2.4 Biochemical Data Matching and Interaction Alerts

The fourth objective is to develop a rule-based lookup engine that cross-references food pH levels and compound profiles in order to detect incompatible food pairings, such as acidic drinks combined with dense proteins, and to flag the resulting digestive reaction risk to the user before the meal is consumed.

3. Literature Survey

A review of recent work relevant to each stage of the NutriTrace pipeline was conducted, spanning ingredient extraction, regulatory harmonisation, risk modelling, and food-interaction analysis. The key studies are summarised below.

S.No	Title	Authors	Year	Techniques Used	Problem Addressed
1	Automated Extraction and Entity Recognition of Food Ingredients from Packaging Images	S. Guru, P. Chaurasia et al.	2025	OCR, Named Entity Recognition (NER), Text Normalization	Inability to parse unstructured, noisy ingredient lists and map obscure E-numbers.
2	Cross-Border Harmonization and Regulatory Mapping of Global Food Additive Databases	M. Chen, R. V. Pellegrino et al.	2024	Relational Schema Mapping, API Ingestion, Multi-DB Querying	Divergent international regulatory safety standards (FDA vs. EFSA vs. FSSAI) for identical food additives.
3	Quantitative Risk Assessment and Dynamic Acceptable Daily Intake (ADI) Modeling	A. Kumar, S. Verma	2025	Statistical Risk Modeling, Weighted Additive Risk Index (ARI)	Lack of personalized additive exposure limits adjusted for user body weight and age constraints.
4	Biochemical Matrix Analysis and Food Combination Interaction Rule Engines	J. R. Navarro-Cabrera et al.	2025	Rule-Based Expert Systems, Automated Knowledge Graph Lookup	Unidentified digestive distress and physiological risks caused by incompatible chemical food pairings.

Taken together, these four studies map directly onto the four stages of the NutriTrace pipeline: entity-level extraction from noisy label text, cross-jurisdictional regulatory reconciliation, personalised statistical risk scoring, and rule-based interaction detection. No single existing work combines all four capabilities into one consumer-facing system, which is the gap NutriTrace is designed to fill.

4. Existing System

Current consumer nutrition platforms, such as Yuka and Open Food Facts, rely primarily on barcode lookups, basic OCR, and static nutrient scoring systems such as Nutri-Score to present basic product information to the user. While useful as a first step toward ingredient transparency, these platforms exhibit several structural limitations that motivate the design of NutriTrace.

These platforms were largely built around a single core idea: match a scanned barcode to a pre-populated product record and surface whatever nutrient information that record contains. This design works well for well-indexed, mainstream products in the region the database was built for, but it breaks down precisely in the situations where chemical-level transparency matters most — imported goods, regional or small-batch products, and any packaging where the barcode is missing, damaged, or simply not yet catalogued. Because the underlying data model was never designed to compare one country's regulatory stance against another's, or to reason about an individual user's physiology, these gaps are structural rather than incidental, and cannot be closed by incrementally adding more barcodes to the same database.

4.1 Key Disadvantages and Gaps

- **Barcode Dependency:** Existing systems fail whenever a product is unindexed in their barcode database or when packaging text is noisy, and they generally lack entity-linking capable of correctly resolving chemical names and E-numbers from free text.
- **Single-Region Bias:** Existing platforms ignore cross-border regulatory differences across the FDA, EFSA, and FSSAI, so an additive that is legal in one country but banned in another is not flagged as a discrepancy.
- **Static, One-Size-Fits-All Scores:** Generic safety flags are used instead of personalised Acceptable Daily Intake (ADI) thresholds that account for the individual user's age and body weight.
- **No Food-Interaction Awareness:** Products are evaluated in isolation, so existing systems fail to detect harmful meal-level pairings, such as high-acid beverages combined with dairy or legumes.
- **No Long-Term Tracking:** Existing platforms lack analytics capable of monitoring a user's cumulative, long-term additive consumption over time, so gradually accumulating exposure risks go unnoticed.

5. Proposed System

NutriTrace is proposed as a full-stack data engine that directly addresses each of the gaps identified in the existing system through a five-layer pipeline: an ingestion and vision layer, a regulatory intelligence layer, a toxicology and risk-scoring layer, a biochemical interaction layer, and a longitudinal tracking and reporting layer.

5.1 System Workflow

The end-to-end workflow begins at the user interface, where a user either uploads a photograph of a food label or scans its barcode. This raw input passes into an ingestion and vision layer, where OCR and multimodal parsing extract ingredient text and structural cues from the image. The extracted text then flows into the core analytics layer, which performs two parallel operations: a regulatory check against the FDA, EFSA, and FSSAI datasets, and an interaction analysis against the biochemical pairing rule base. Results from both operations, together with the user's personalised ADI calculation, are written to a central database. Finally, a dashboard layer renders safety indices, warnings, and downloadable reports back to the user.

This flow was deliberately structured as a set of loosely coupled stages rather than a single monolithic process, so that each stage — vision, regulatory mapping, risk scoring, interaction detection, and reporting — can be developed, tested, and scaled independently. The central database acts as the integration point between stages, meaning that a given scan's structured ingredient list, regulatory findings, computed ARI/ADI values, interaction flags, and timestamp are all persisted together, and can subsequently be replayed into the monthly tracking module without re-running the vision or regulatory layers.

5.2 Core Modules

- **Ingestion and Vision Layer:** Responsible for OCR-based text extraction and multimodal image processing of food packaging, converting an unstructured photograph into a structured ingredient list.
- **Regulatory Intelligence Layer:** Cross-references each extracted additive against the FDA, EFSA, and FSSAI datasets in parallel and surfaces any jurisdictional discrepancy in legal status or permitted usage limit.
- **Toxicology and Risk-Scoring Layer:** Computes the Additive Risk Index (ARI) and the personalised Acceptable Daily Intake (ADI) using WHO/JECFA thresholds combined with the user's age and body weight.
- **Biochemical Interaction Layer:** Applies a rule-based expert system over food pH levels and compound profiles to detect and flag incompatible food pairings before consumption.
- **Longitudinal Tracking Layer:** Aggregates additive exposure on a monthly basis, maintaining a cumulative record that supports early warnings once exposure patterns begin to trend toward risk.
- **Dashboard and Reporting Layer:** Presents safety indices, warnings, and downloadable reports to the user through an accessible, consumer-facing interface.

5.3 Advantages Over the Existing System

- Extracts chemical additive information directly from packaging images rather than depending solely on a barcode match.
- Compares additive safety simultaneously across FDA, EFSA, and FSSAI standards instead of relying on a single regional database.
- Computes a personalised ADI and ARI based on the individual user's age and body weight, rather than a generic score.
- Detects harmful food-pairing interactions at the meal level, not just at the level of a single product.
- Maintains a cumulative, longitudinal record of additive exposure, enabling early warnings that existing static tools cannot provide.

6. System Requirements

6.1 Functional Requirements

- The system shall accept a photographed food label as input and extract all visible ingredient text using OCR.
- The system shall normalise extracted ingredient names and E-numbers into a canonical additive identifier.
- The system shall query FDA, EFSA, and FSSAI datasets for each identified additive and report jurisdictional differences.
- The system shall compute a personalised Acceptable Daily Intake and Additive Risk Index using the user's age and body weight.
- The system shall check newly scanned items against previously logged items for known biochemical interaction risks.
- The system shall maintain a persistent, per-user monthly log of cumulative additive exposure.
- The system shall present results through a dashboard summarising safety indices, warnings, and downloadable reports.

6.2 Non-Functional Requirements

- **Accuracy:** OCR and entity-recognition components must achieve high extraction accuracy on standard packaging fonts, including partially obscured or curved text.
- **Scalability:** The regulatory data layer must scale to accommodate additional jurisdictions and an expanding additive knowledge base without redesign.
- **Performance:** End-to-end label-to-dashboard latency should remain low enough for a practical, in-store scanning experience.
- **Data Privacy:** User health attributes such as age and body weight must be stored securely and used only for on-device or authorised server-side ADI computation.
- **Extensibility:** The biochemical rule engine and regulatory schema must be designed so that new rules and jurisdictions can be added as configuration rather than code changes.

7. Technology Stack

The NutriTrace platform is designed as a modular, full-stack data engineering system. The technology choices for each layer are summarised in the table below.

Layer	Representative Technologies	Purpose
Ingestion & Vision	OCR engines, multimodal image parsing, NLP-based NER	Convert packaging images into structured ingredient text
Regulatory Intelligence	Relational database, schema mapping, REST API ingestion	Unify FDA, EFSA, and FSSAI datasets under one additive schema
Risk Scoring	Statistical modelling libraries, weighted scoring functions	Compute the Additive Risk Index and personalised ADI
Interaction Engine	Rule-based expert system, knowledge graph lookup	Detect harmful food pairings from pH and compound profiles
Storage & Tracking	Relational / document database for per-user exposure logs	Maintain cumulative monthly additive exposure records
Dashboard & Reporting	Web-based front end, charting library, PDF report export	Present safety indices, warnings, and downloadable reports

8. Detailed Module Description

8.1 Ingestion and Vision Module

The ingestion module is the entry point for all user data. It accepts an uploaded or captured image of a food label and applies an OCR pass tuned for the small, dense typography typical of ingredient panels. Because ingredient text is frequently printed on curved packaging, in low contrast, or in a non-Latin script, the module applies pre-processing steps such as perspective correction and contrast enhancement before text recognition. The raw OCR output — which is often noisy and fragmented — is then passed through an NLP normalisation stage that resolves spelling variants, abbreviations, and regional naming conventions into a single canonical additive identifier, and maps bracketed E-numbers directly onto that identifier.

8.2 Regulatory Intelligence Module

Once a canonical list of additives has been produced, the regulatory intelligence module queries a unified schema built from FDA, EFSA, and FSSAI source datasets. For each additive, the module retrieves the maximum permitted usage level in each jurisdiction, the categories of food in which it may be used, and any outright bans. Discrepancies — for example, an additive permitted without restriction in one jurisdiction but banned in another — are flagged explicitly, giving the user a consolidated, side-by-side view of global regulatory opinion rather than a single-country answer.

8.3 Toxicology and Risk-Scoring Module

This module computes two related but distinct outputs. The Additive Risk Index (ARI) is a weighted composite score reflecting the severity of known health concerns associated with an additive, the number of jurisdictions restricting it, and the concentration detected in the scanned product. The Acceptable Daily Intake (ADI) calculation instead answers a more specific question — how much of this additive can this particular user safely consume in a day — by combining the WHO/JECFA reference dose per kilogram of body weight with the user's declared age and weight. Together, the ARI and ADI give the user both a general risk signal and a personalised consumption guideline.

8.4 Biochemical Interaction Module

The interaction module maintains a knowledge base of food pH values and compound profiles, together with a set of rules describing known adverse pairings — for instance, the combination of a highly acidic beverage with a protein-dense food, which can cause rapid protein curdling and consequent digestive distress. When a user logs multiple items within a short time window, the module checks each new item against recently logged items and raises an alert if a known incompatible pairing is detected.

8.5 Longitudinal Tracking and Dashboard Module

Every scan a user performs is written to a persistent, per-user record. The tracking module aggregates this record on a monthly basis, computing cumulative exposure to each additive class and comparing it against safe long-term thresholds. The dashboard module then renders this information — current-scan safety indices, jurisdictional warnings, interaction alerts, and the monthly exposure trend — in a single consumer-facing interface, and allows the user to export a downloadable report summarising their additive exposure history.

9. Risk Assessment Framework

The statistical core of NutriTrace rests on two complementary calculations: the Additive Risk Index and the personalised Acceptable Daily Intake.

9.1 Additive Risk Index (ARI)

The ARI is computed as a weighted combination of three factors: the severity of documented health concerns associated with the additive, drawn from the toxicological literature; the breadth of regulatory restriction, measured as the proportion of the tracked jurisdictions (FDA, EFSA, FSSAI) that restrict or ban the additive; and the concentration of the additive detected in the specific product being scanned, relative to its typical usage range. Each factor is normalised to a common scale before being combined, so that the resulting ARI is comparable across additives of very different chemical types.

9.2 Personalised Acceptable Daily Intake (ADI)

The ADI calculation follows WHO/JECFA methodology, under which a reference dose is expressed in milligrams of additive per kilogram of body weight per day. NutriTrace personalises this reference dose using the user's declared body weight to produce an absolute daily limit in milligrams, and further adjusts the result using age-based modifiers reflecting the different metabolic and physiological sensitivities of children, adults, and older users. The system then compares the additive quantity detected in a scanned product against this personalised limit, expressing the result as a percentage of the user's daily allowance consumed by that single item.

10. Methodology and Implementation Plan

6.1 Data Sources

The system draws on publicly available regulatory datasets from the FDA, EFSA, and FSSAI for additive legality and usage limits, WHO/JECFA technical reports for toxicological reference thresholds, and an internal biochemical compound and pH reference table constructed for the food-interaction rule engine. Ingredient and packaging data is captured directly from user-submitted images at run time.

6.2 Technical Approach

- **Vision and Language Processing:** OCR is applied to the captured packaging image, followed by Named Entity Recognition and text normalisation to resolve ingredient names, synonyms, and E-numbers into canonical identifiers.
- **Regulatory Schema Mapping:** A relational schema unifies the FDA, EFSA, and FSSAI datasets under a common additive identifier, allowing a single query to retrieve each jurisdiction's legal status and usage limit for a given additive.
- **Statistical Risk Modeling:** A weighted Additive Risk Index is computed per additive, and the Acceptable Daily Intake is calculated dynamically from the user's age and body weight using WHO/JECFA reference formulas.
- **Rule-Based Interaction Engine:** A knowledge-graph-backed, rule-based expert system cross-references pH and compound profiles to flag interactions such as acidic beverages paired with dense proteins, which are known to cause rapid protein curdling and digestive distress.

6.3 Evaluation Plan

The completed system will be evaluated along three dimensions: extraction accuracy, measured by comparing OCR/NER output against manually annotated ingredient labels; regulatory coverage, measured by the proportion of common additives correctly mapped across all three jurisdictions; and interaction-detection precision, measured against a curated set of known problematic food pairings drawn from the literature reviewed in Section 3.

11. Testing Strategy and Project Timeline

11.1 Testing Strategy

- **Unit Testing:** Individual functions within the OCR normalisation, ARI computation, and interaction rule engine are tested against known inputs and expected outputs.

- **Integration Testing:** The full pipeline is exercised end-to-end, from image upload through to dashboard rendering, to confirm that data passed between the vision, regulatory, risk-scoring, and interaction layers remains consistent.
- **Regulatory Validation:** Sample additives with known FDA, EFSA, and FSSAI statuses are used to confirm that the regulatory intelligence layer surfaces the correct jurisdictional discrepancies.
- **User Acceptance Testing:** A small group of representative users tests the label-scanning workflow under real packaging conditions to validate OCR accuracy and dashboard usability.

11.2 Project Timeline

Phase	Key Activities
Phase 1 — Requirements & Design	Problem definition, literature survey, requirement specification, architecture design.
Phase 2 — Ingestion & Vision	OCR pipeline development, NLP normalisation, canonical additive identifier mapping.
Phase 3 — Regulatory & Risk Engine	FDA/EFSA/FSSAI schema unification, ARI and ADI computation modules.
Phase 4 — Interaction & Tracking	Biochemical rule engine, monthly exposure tracker, database integration.
Phase 5 — Dashboard & Testing	Dashboard development, unit/integration/user-acceptance testing, final report.

11.3 Ethical and Data Privacy Considerations

Because NutriTrace processes personal health attributes such as age and body weight in order to compute a personalised ADI, the system design treats this data as sensitive by default. User health attributes are collected only where strictly necessary for the risk calculation, are not shared with third parties, and are stored separately from any identifying account information wherever the architecture allows it. The regulatory and toxicological reference data used by the system is drawn exclusively from published FDA, EFSA, FSSAI, and WHO/JECFA sources, and the system is explicit that its output is intended to inform, not replace, professional medical or dietary advice.

12. Conclusion

NutriTrace bridges the gap between raw, unstructured food supply chain data and personalised consumer health analytics. By combining an AI-driven, multi-modal ingestion pipeline with multi-jurisdictional regulatory intelligence, the system is capable of parsing complex ingredient lists and evaluating the regulatory discrepancies that arise between different national food safety authorities. The addition of a biochemical food-interaction engine and a monthly cumulative exposure tracker allows NutriTrace to address both acute digestive risks and long-term toxicological build-up — two categories of harm that existing consumer nutrition platforms largely overlook.

Taken as a whole, this project demonstrates the high-impact potential of combining data engineering with biochemical analytics, delivering a scalable, full-stack framework that can support consumers, health researchers, and regulatory bodies in making more informed, data-driven decisions about global food safety.

13. Future Enhancements

- Extending the regulatory intelligence layer to cover additional jurisdictions beyond the FDA, EFSA, and FSSAI, such as Codex Alimentarius international standards.
- Improving OCR robustness for damaged, curved, or low-light packaging images using fine-tuned vision-language models.
- Expanding the biochemical interaction rule base with a larger, continuously updated compound and pH reference dataset.
- Adding mobile-first barcode scanning as a complementary input path alongside image-based label capture.
- Introducing personalised dietary recommendations that combine ADI tracking with the user's broader health profile.

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